Coasters and pendulums

antocciphysics | TEACHER KEY | CPA

Learning target: Use height differences to predict speed in ideal motion.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 9; 3 course days later.*
ΔU=+12 J; gained. Less negative means greater energy.

02. Explain a worked example (worked)
U converts to K then back to U; equal ideal turning heights, zero speed at each turning point.

03. Look and explain (model)
Same final speed if no rotation or losses; times can differ.

04. Build the solution (practice)
60 J lost U, 60 J gained K.

05. Use the idea (reason)
h is vertical height difference, not distance along track; measure perpendicular to level reference.

06. Choose a method (mixed) *Review: Unit 3, Day 14; 12 course days later.*
24 m. The constant-Δv straight-line model omits full orbital dynamics.

07. Exit check (exit)
12 J.

08. STRETCH: Apply it another way (stretch)
Initial 8 J, final 6 J, loss 2 J to thermal/sound/etc.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 9; 3 course days later.*
U_i=-35 J; U_f=-15 J; ΔU=+20 J; gravity work=-20 J.

02. Explain a worked example (worked)
U converts to K then back to U; equal ideal turning heights, zero speed at each turning point.

03. Represent and interpret (model)
Same final speed if no rotation or losses; times can differ.

04. Solve and justify (practice)
v^2=36+2(10)(5)=136; v 11.7 m/s.

05. Analyze the reasoning (reason)
h is vertical height difference, not distance along track; measure perpendicular to level reference.

06. Choose a method (mixed) *Review: Unit 3, Day 14; 12 course days later.*
J=12 N·s; v_f≈0 m/s from mv_i+J=mv_f.

07. Exit check (exit)
No, not from rest and without input; would require negative K at 6 m.

08. STRETCH: Apply it another way (stretch)
Top height 4; v^2=80, v≈8.94. N/m=v^2/R-g=30 N/kg, positive contact.